

Pre-Test/Post-Test to Measure Student Learning during the Flag Maker Unplugged PDC Activity

* Indicates required question

1. Enter your HPU email *

Answer the following questions to the best of your knowledge.

2. **Task Decomposition:** Which of the following best describes task decomposition? *

Mark only one oval.

- ☐ The method of organizing tasks in a sequential manner.
- ☐ The process of breaking down a large task into smaller, independent tasks that can be executed concurrently.
- ☐ The technique of reducing the number of tasks to improve performance.
- ☐ The strategy of assigning tasks to a single processor.

3. **Speedup:** Speedup is defined as the ratio of the time taken to solve a problem on a single processor to the time taken on a parallel system. *

Mark only one oval.

- ☐ True
- ☐ False

4. **Contention:** What is contention in parallel computing? *

Mark only one oval.

- ☐ The process of dividing a task into smaller sub-tasks.
- ☐ The increase in computational speed by adding more processors.
- ☐ The competition between multiple processors for shared resources.
- ☐ The ability of a system to handle a growing amount of work.

5. **Scalability:** Scalability refers to the ability of a parallel system to increase its performance proportionally with the addition of more processors. *

Mark only one oval.

- ☐ True
- ☐ False

6. **Pipelining:** What is pipelining in the context of parallel computing? *

Mark only one oval.

- ☐ The process of executing multiple tasks simultaneously.
- ☐ The technique of overlapping the execution of multiple instructions to improve performance.
- ☐ The method of dividing a task into smaller sub-tasks.
- ☐ The strategy of reducing contention among processors.

